Experiment No : 10

Date:

# Dipole Antenna

## Aim

Design and Simulate a Dipole Antenna of resonant frequency, $f_r$=700MHz.

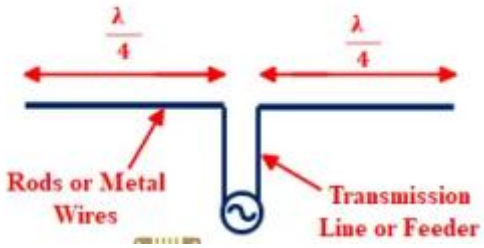
## Theory

A dipole antenna (also known as a doublet or dipole aerial) is defined as a type of RF (Radio Frequency) antenna, consisting of two conductive elements such as rods or wires. The dipole is any one of the varieties of antenna that produce a radiation pattern approximating that of an elementary electric dipole. Dipole antennas are the simplest and most widely used type of antenna.

A 'dipole' means 'two poles' hence the dipole antenna consists of two identical conductive elements such as rods or metal wires. The length of the metal wires is approximately half of the maximum wavelength (i.e.,$\lambda/2$) in free space at the frequency of operation.

This wire or rod is split at the centre, and the two sections are separated by an insulator, these sections are known as an antenna section.

These two antenna sections are connected to a feeder or coaxial cable at the end closest to the centre of the antenna. Note that wavelength is the distance between two consecutive maximum or minimum points. The basic dipole antenna with the centre feed point is shown in the figure below.

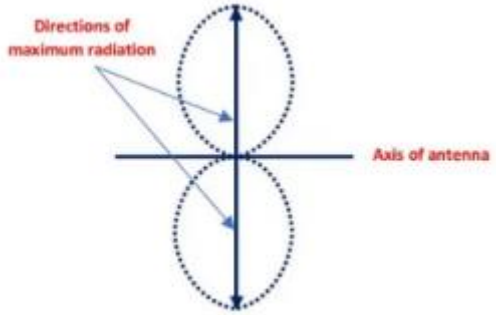


The radio-frequency (RF) voltage source is applied to the center between the two sections of the dipole antenna. This voltage and a current flowing through the two conductive elements produce a radio signal or an electromagnetic wave to be radiated outwards from the antenna.

The current is maximum and voltage is minimum at the center of the dipole antenna. Conversely, the current is minimum and voltage is maximum at the ends of the dipole antenna.

The radiation pattern of the basic dipole antenna is shown in the figure below. It is perpendicular to the axis of the antenna.

Note that the radiation pattern is the graphical representation of the radiation properties of the antennas as a function of space i.e., the radiation pattern of the antenna describes how the antenna radiates energy out into space.




The dipole antenna is one type of transducer which converts electrical signals into RF electromagnetic waves and radiates them at the transmitting side and converts RF electromagnetic waves into electrical signals at the receiving side.

**Sample design**

We can design many dipole antennas that will operate on the HF (high frequency), VHF (very high frequency), and UHF (ultra-high frequency) bands of the radio frequency spectrum.
Let’s design a 1 MHZ dipole antenna.

Selection of Length of the Dipole Antenna
As we know that the wavelength of a radio wave or any other wave varies inversely proportional to the frequency. it is given by:

$\lambda \ \alpha \ 1/f$

$\lambda = c/f$

Where, C = velocity of light = $3*10^8$ m/s

f = frequency, in Hertz

λ= wavelength, in meter

Thus,

$\lambda = 3*10^8 / f(Hz)$ meters

Now, at a half-wavelength, the length of the antenna is given by,

$\lambda/2 = 150*3.28/f(MHz) = 492/f(MHz)$ feet

Thus, from equation (3) we can say that, if we used a 1 MHz radio transmitter, then the basic length of the antenna wire will be 150 meters or 492 feet or 5905 Inches.

This is correct if we neglect the “end effect”. This “end effect” is the dielectric effect of the air at the end of the antenna that increases the effective length of the antenna. Due to the end effect, an antenna wire acts as 5% longer than the actual length. This will produce interference between the exciting and oscillating currents and due to that oscillation amplitude may be weakened.

Hence, to counterbalance the “end effect” and to make the antenna works properly, it is necessary to cut the antenna wire about to 5% and makes its physical length approximately 95% of half of the wavelength.

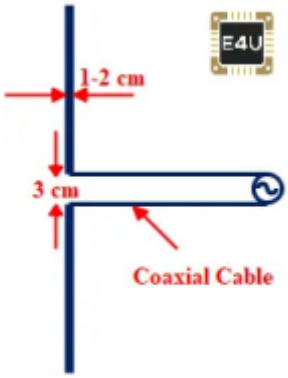
Thus, to get the practical length of the antenna wire, the value multiplied by a factor K to the basic length of the antenna wire, i.e.,

$\lambda = 468/f(MHz)$ feet

The value of K depends on the thickness of the conductor and the operating frequency. This value of K is accurate for antenna wire at a frequency of up to 30 MHz.

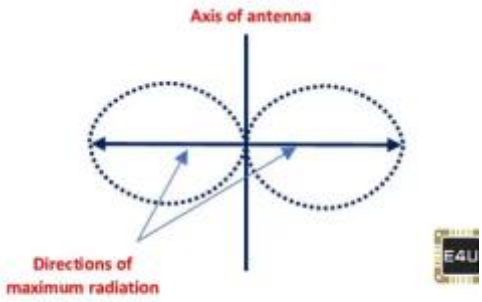
FM Dipole Antenna

FM dipole antenna is defined as the vertically polarized half-wave half dipole antenna. The FM dipole antenna is mostly used as it is to build and it is suitable to provide improved reception of VHF FM broadcast. The FM dipole antenna is shown in the figure below.




FM Dipole Antenna

The radiation pattern of the FM dipole antenna is perpendicular to the axis of the antenna i.e., It is horizontal because the FM dipole antenna is a vertically polarized half-wave dipole antenna. The radiation pattern of the FM dipole antenna is shown in the below figure.




Radiation Pattern of a FM Dipole Antenna

**Procedure**

1. Creating the project
2. Draw the cylinder(dipole arm). Right click on the **CreateCylinder** to change the dimensions and position of the cylinder

- Centre position : 0,0,1 mm
- Axis : z
- Radius : 1 mm
- Height : 107 mm

**3. Assign material**

➔ the cylinder
➔ Right click on the cylinder
➔ Click on Assign Material
➔ Choose perfect conductorSelect

4. Draw the cylinder(dipole arm). Right click on the **CreateCylinder** to change the dimensions and position of the cylinder.

- Centre position : 0,0,-1 mm
- Axis : Z
- Radius : 1 mm
- Height : -107 mm

**5. Assign material**

➔ Select the cylinder
➔ Right click on the cylinder
➔ Click on Assign Material
➔ Choose perfect conductor

6. Draw the rectangle. Right click on the **CreateRectangle** to change the dimensions and position of the rectangle.

- Position : 0,-1,-1 mm
- Axis : X
- Ysize : 2 mm
- ZSize : 2 mm

7. **Select port**

➔ Select the rectangle
➔ Right click on the rectangle
➔ Click on Assign Excitation
➔ Choose Lumped Port
➔ Select new line
➔ Draw a line from bottom middle to top middle of the rectangle
➔ Click next
➔ Click finish

8. Draw the Radiation Cylinder

9. Right click on the **CreateCylinder** to change the dimensions and position of the cylinder.

- Centre position : 0,0,-200 mm
- Axis : Z
- Radius : 100 mm
- Height : 400 mm

10. **Assign Material**

Select cylinder

Right click on cylinder

Click on Assign material

Select Air

Click on Ok

11. **Radiation**

➔ Select cylinder
➔ Right click on the cylinder
➔ Click on Assign boundary
➔ Click on Radiation
➔ Choose Radiating Only

➔ Click on Ok

**12. Solution setup**

➔ Go to Analysis
➔ Click on Add solution setup

- Solution frequency : 0.7 GHz
- Number of passes : 10
- Click on Ok

13. **Analysis setup 1**

➔ Go to setup 1
➔ Click on Add frequency sweep

- Sweep type : fast
- Start : 1 MHz
- Stop : 1.5 GHz
- Step size : 0.01

**14. Validate**

**15. Analyze all**

**Design and Calculations**

Given f = 700MHz=0.7GHz

$c = f \lambda$

where $c = 3*10^8$ m/s

$\lambda = c/f = 3*10^8 / 0.7*10^9$

$= 428.5$ mm

Length of the antenna = $\lambda /4$

$= 107.125$ mm

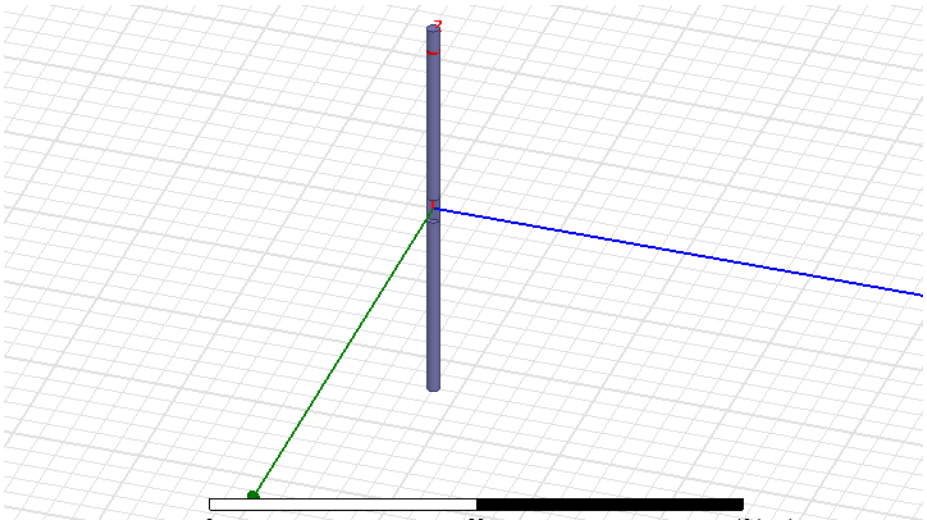
Dimension of radiation cylinder :

Height and Radius of the radiation cylinder measured from the end points of physical dimensions of the antenna $> = 2D^2/\lambda$

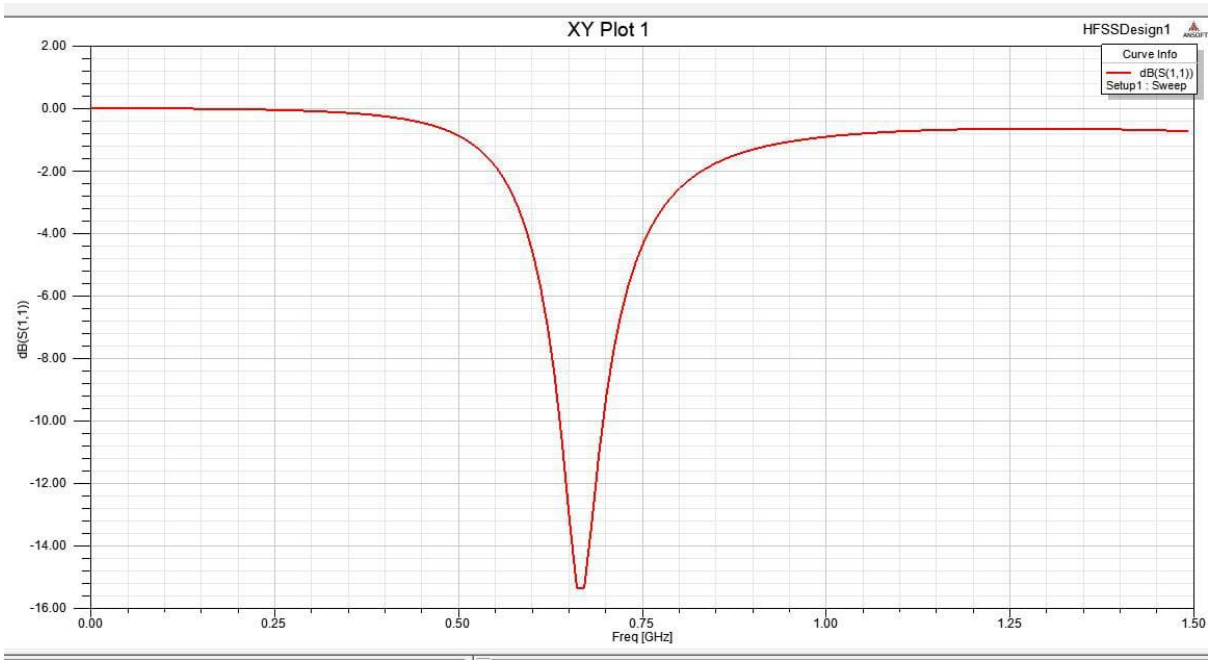
where $D = \lambda/4$

radius, r >= 53.5625 mm

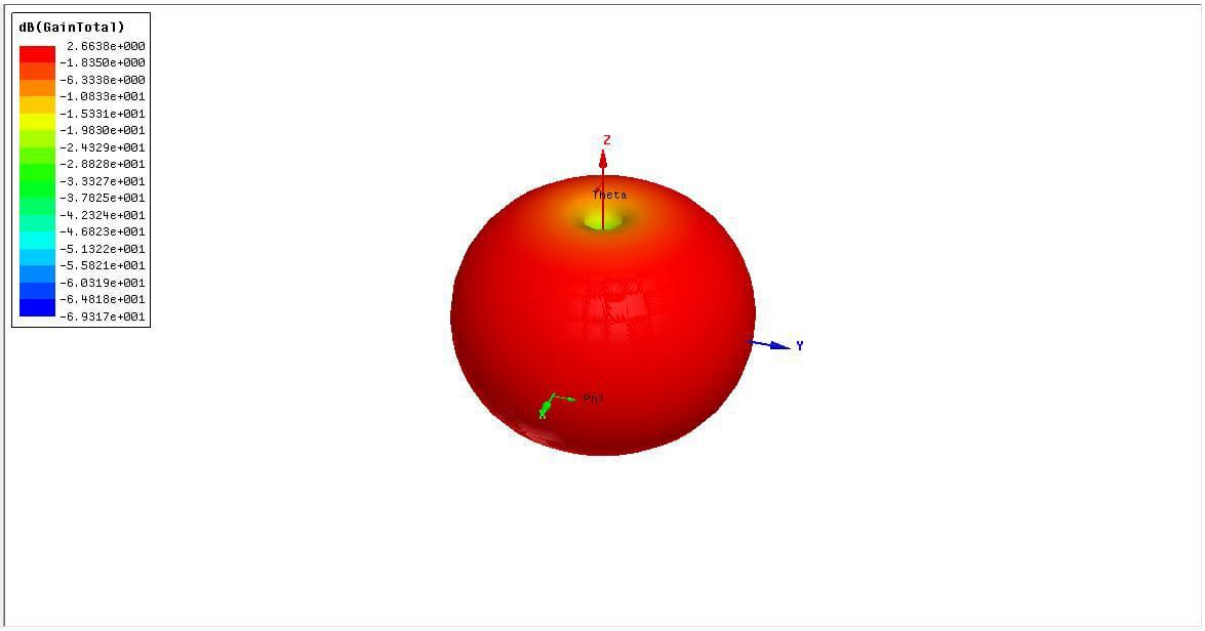
## Observation



## Return Loss




## 3D Polar Plot




## Result

Designed and simulated a dipole antenna with resonant frequency 0.7 GHz.